



INT8 MatVec RTL Optimization

Verified VTR 45 nm implementation evidence

8.3230% better

paired composite area, delay and active-power estimate

15.5943% area

3.1868% delay

5.7081% active power

Claim boundary. Academic VTR PTM 45 nm post-route estimates at 0.9 V and 85 degrees C. This is not commercial-FPGA characterization, Vivado, ASIC signoff, a board measurement, measured power or silicon evidence.

1. Executive result

WHAT CHANGED AND WHAT IS SUPPORTED

Evolther optimized one bounded hardware primitive: a combinational 4x4 signed-INT8 matrix-vector datapath with four signed-INT32 outputs. The frozen interface, mathematical dot product and observable outputs did not change.

Primary metric	Baseline	Optimized	Paired improvement
Total area estimate	33,892,339	28,607,064	15.5943%
Critical-path delay	11.1642 ns	10.8085 ns	3.1868%
Active total power	22.4567 mW	21.1749 mW	5.7081%
Composite score	1.000000	0.916770	8.3230%

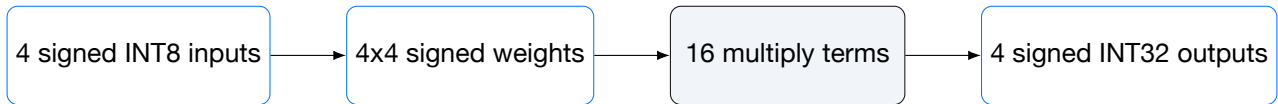
The accepted implementation replaces unnecessary 32-bit extensions before accumulation with a balanced 17/18-bit adder tree. This reduces mapped width and depth without changing any product or dot-product result.

Acceptance evidence. 151 deterministic simulations pass, exhaustive combinational equivalence covers every 160-bit input assignment, and 64 held-out implementation pairs satisfy the fixed statistical rule. Baseline and optimized records reproduce exactly.

2. Why this circuit matters

A SMALL BUT RELEVANT AI-HARDWARE KERNEL

Quantized linear, projection and convolution operators reduce to repeated dot products. This case isolates that arithmetic core: four signed-INT8 activations multiplied by sixteen signed-INT8 weights to produce four exact signed-INT32 sums.



The largest four-term magnitude is 65,536, well inside signed INT32. The optimized 17/18-bit intermediate widths are therefore range-correct for the frozen 4x4 contract.

This is a proof point for a central AI-hardware primitive, not a claim about a complete layer, neural network, memory hierarchy, accelerator platform or deployment.

3. The RTL transformation

NARROWER AND BALANCED ACCUMULATION

The baseline sign-extends every 16-bit product to 32 bits and then expresses four wide additions per output. The optimized version first adds product pairs at 17 bits, then combines the two pair sums at 18 bits and finally sign-extends the exact result.

Frozen baseline	Optimized RTL
16 products	16 identical products
16 separate 32-bit extensions	No full-width product-extension bank
Four wide accumulation expressions	Balanced two-level adder trees
32-bit internal add operands	Range-correct 17/18-bit intermediates

```
// Optimized structure for one output row
s00 = {p00[15], p00} + {p01[15], p01};
s01 = {p02[15], p02} + {p03[15], p03};
d0  = {s00[16], s00} + {s01[16], s01};
y0  = {{14{d0[17]}}, d0};
```

The exact public before/after patch is shipped with the case. No interface, protocol, arithmetic convention or output width is part of the optimization freedom.

4. Correctness before PPA

FAIL CLOSED VERIFICATION

The candidate reaches PPA scoring only after all earlier gates pass:

1. SystemVerilog structure and synthesis safety.
2. 151 deterministic golden simulations, including signed extremes, individual lanes, individual rows and seeded-random vectors.
3. Unbounded Yosys combinational equivalence of all four outputs for every possible 160-bit input assignment.
4. Width-sensitive synthesis and implementation checks that reject unsupported or oversized multiplier structures.
5. VTR route, timing and power extraction for every required pair.

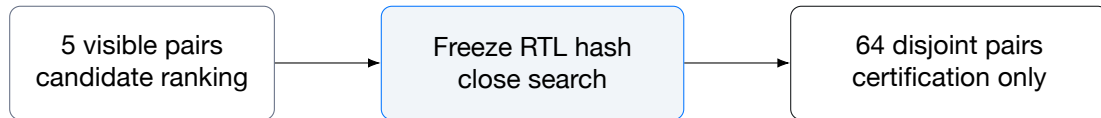
An inconclusive formal result, missing metric, timeout, route failure or functional mismatch is invalid. It cannot receive an improvement score.

Gate	Scope	Result
Golden simulation	151 deterministic signed/extreme cases	PASS
Formal equivalence	Every 160-bit input assignment	PASS
PPA validity	Resource, multiplier-width and extraction limits	PASS
Independent replay	Baseline and optimized paired records	EXACT

5. Measurement and certification

SEARCH EVIDENCE IS NOT ACCEPTANCE EVIDENCE

The implementation target matches the academic VTR/PTM operating point used by the SHA-1 methodology case: a homogeneous LUT6 FPGA architecture, PTM 45 nm model, 0.9 V, 85 degrees C and a pinned Linux/amd64 toolchain image.



The score is the paired geometric mean of total area, critical-path delay and active-total-power ratios. Lower is better. Acceptance is fixed before measurement: the composite one-sided 95% upper confidence bound must be below 1.0, while no primary metric may have a one-sided 95% lower bound above 1.0.

The public compact certificate retains all 64 baseline/optimized metric pairs under blinded pair labels. It does not publish the optimization implementation, private execution data or reserved placement identities.

6. Certified result

PAIRED ESTIMATES OVER THE FIXED HELD-OUT SAMPLE

Metric	Estimate	Two-sided 95% interval	Wins / losses
Area improvement	15.5943%	15.5355% to 15.6531%	64 / 0
Delay improvement	3.1868%	2.5843% to 3.7855%	54 / 10
Active-power improvement	5.7081%	5.1622% to 6.2509%	64 / 0
Composite improvement	8.3230%	8.1997% to 8.4462%	64 / 0

Area and active power improve in all 64 pairs. Critical-path delay improves in 54 and regresses in 10, while its paired estimate and non-regression bound still pass. The composite improves in every held-out pair.

Interpretation. The composite estimate is a target-specific modeled comparison. It supports a reproducible result on the frozen academic architecture, not a universal 8.3230% improvement for every device, toolchain or workload.

7. Public trust boundary

WHAT A CLIENT CAN INSPECT

The public case includes the baseline and optimized RTL, exact patch, compact paired certificate, checksums, this report and a fail-closed Python verifier. From the case directory:

```
python3 verify.py
```

The verifier checks file identities, the exact RTL patch, schema strictness, all paired values, derived area/power/frequency relationships, geometric aggregates, confidence intervals, acceptance bounds, resource limits and independent-replay attestations.

Public evidence	Outside the public package
Before/after RTL and patch	Optimization implementation
Paired compact measurements	Private execution data
Correctness and replay attestations	Internal search history
Tool and target identities	Future reserved certification pools

Baseline RTL SHA-256
c2472728e2d52b29ad451f5c9779ebb749c318e9be048d03e6a6555c735eda53

Optimized RTL SHA-256
25327189f78f238e909fa5940afff9e2b000547336cdc614ee9ea9c4633aa11f

Compact certificate SHA-256
e77ed39c79559af8cbbd2efea4fb67045740684ac4fc35699213c3cc1858906

Commercial next step. A customer pilot should freeze one owned design interface, define success and validity limits before search, select an implementation target and agree which evidence may be published. Results remain subject to that target and contract.